

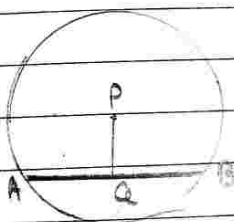
17. Circle - Chord and Arc

Properties of chord of a circle:-

- 1] The perpendicular drawn from the centre of a circle to its chord bisects the chord.
- 2] The segment joining the centre of a circle and midpoint of its chord is perpendicular to the chord.

Practice Set 17.1

- 1] In a circle with centre P, chord AB is drawn of length 13 cm. Seg PQ $\perp$ chord AB, then find $l(QB)$.



→ Chord $AB = 13$ cm.

Seg PQ $\perp$ chord AB.

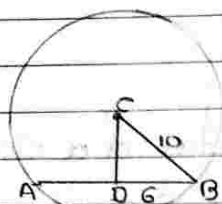
Perpendicular drawn from the centre of a circle to its chord bisect the chord.

$$\therefore l(QB) = \frac{AB}{2}$$

$$= \frac{13}{2}$$

$$l(QB) = 6.5 \text{ cm}$$

- 2] C is the centre of the circle whose radius is 10 cm. Find the distance of the chord from the centre if the length of the chord ~~is~~ is 12 cm.



→ In circle 'C' is the centre.

seg CB is a radius.

$$CB = 10 \text{ cm}$$

$$DB = 6 \text{ cm}$$

Here,

seg CD $\perp$ chord AB

By property of chord,

$$CD = \frac{AB}{2}$$

$$= \frac{12}{2}$$

$$CD = 6 \text{ cm}$$

Now,

In $\triangle CDB$

By pythagoras the^m

$$CB^2 = CD^2 + DB^2$$

$$10^2 = CD^2 + 6^2$$

$$100 = CD^2 + 36$$

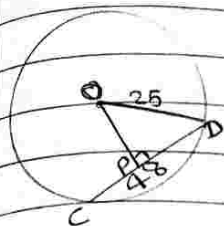
$$CD^2 = 100 - 36$$

$$CD^2 = 64$$

$$CD = 8 \text{ cm}$$

Distance of chord from the centre is 8 cm.

- 2] Radius of a circle with centre O is 25 cm. Find the distance of a chord from the centre if length of the chord is 48 cm.



→ In circle 'O' is the centre.
seg OD is a radius.

$$OD = 25$$

$$CP = 48$$

Hence,

seg OP $\perp$ chord CD
By property of chord,
 $CP = \frac{CD}{2}$

$$CP = 24 \text{ cm}$$

Now,

In ΔOPD
By Pythagoras theorem

$$OD^2 = OP^2 + PD^2$$

$$25^2 = OP^2 + 24^2$$

$$625 = OP^2 + 576$$

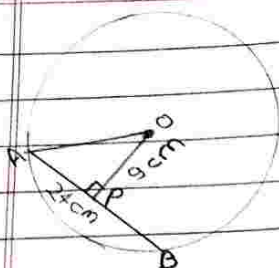
$$OP^2 = 625 - 576$$

$$OP^2 = 49$$

$$OP = 7 \text{ cm}$$

Distance of chord from the centre is 8 cm.

3. O is centre of the circle. Find the length of radius, if the chord of length 24 cm is at a distance of 9 cm from the centre of the circle.



→ In 'O' is centre.

Seg AO is radius.

seg OP is chord.

$$OP = 9 \text{ cm}$$

$$AP = 12$$

$$AP = \frac{AB}{2}$$

$$= \frac{24}{2}$$

$$AP = 12 \text{ cm}$$

Now,

$$AO^2 = OP^2 + AP^2$$

$$AO^2 = 9^2 + 12^2$$

$$AO^2 = 81 + 144$$

$$AO^2 = 225$$

$$AO = 15 \text{ cm}$$

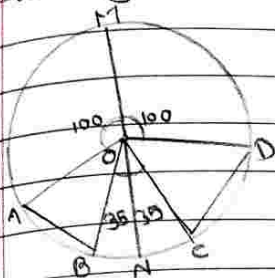
Arcs corresponding to the chord of a circle

Congruent arcs

If the measures of two arcs of circle are same then two arcs are congruent.

Practice Set 17.2

2. In the adjoining figure O is the centre of the circle whose diameter is MN . Measures of some central angles are given in the figure. Hence find the following



1] $m\angle AOB$ and $m\angle COD$.

→ Sum of measure of circle = 360°

$$100^\circ + 100^\circ + 35^\circ + 35^\circ + m\angle AOB + m\angle COD = 360$$

$$270^\circ + m\angle AOB + m\angle COD = 360$$

$$m\angle AOB + m\angle COD = 360 - 270$$

$$m\angle AOB + m\angle COD = 90^\circ$$

But,

$$m\angle AOB \cong m\angle COD$$

$$\therefore m\angle AOB \cong m\angle COD = 45^\circ$$

2] Show that $\text{arc } AB \cong \text{arc } CD$

→ The central angles of each arc are equal

$$(\text{i.e.}) = 45^\circ$$

$$\therefore \text{Arc } AB \cong \text{Arc } CD$$

3] Show that chord $AB \cong$ chord CD

→ The chords related to congruent arcs are congruent

$$\therefore \text{chord } AB \cong \text{chord } CD$$

1. The diameters PQ and RS of the circle with center C are perpendicular to each other at C . State, why are PS and SR are ~~also~~ congruent write the other arcs which are congruent to arc PS .

14.

① Si

② Co